

5- SYSTICK

1. Introduction - Peripherals
2. Timer
3. PWM – Pulse Width Modulations
4. GPIO
5. Low-power Drivers (LED, Relay)
6. Power Drivers (DC Motor)

5.1 - PERIPHERALS

A Microprocessor-based system consists of three types of components:

1. **microprocessor** (the **core** of the MCU) - executes the instructions of a program;
2. **memory** - store code and data;
3. **peripherals** - perform specific functions particularly related to I/O. Peripherals provide the means for the system to sense, actuate and communicate with the world.

A microprocessor-based system without peripherals would be unable to interact with the rest of the world !

PERIPHERALS

Classes of peripherals in a microprocessor-based system:

- Digital I/O: input/output of digital signals,
- Analog I/O: input/output of analog signals,
- Timing: pulse generation, pulse measurement, PWM, ...
- Storage: non-volatile memory, file system, ...
- Communications: RS-232, SPI, I2C, CAN, USB, Ethernet, ...
- HMI: touch-screen, keyboard, ...
- Imaging: camera interface, ...
- System management: clock generation, watchdog, power management, ...

5.2 TIMERS/COUNTERS

Timers/Counters are essential components of a Microcontroller Unit (MCU). They consists of a **digital counter circuit** that counts pulses on their input. **Timers** count clock pulses and **Counters** count pulses of a signal present on their input pin.

Timers/Counters are used for:

- Measuring time, particularly time intervals,
- Counting events,
- Keeping track of current date and time (Real-Time Clock - RTC),
- Generating digital waveforms (PWM - Pulse-Width Modulation),
- Generating periodic interrupts to the MCU (Operating System Clock),
- Generating periodic signals to other peripheral, such as the command to start an Analog-to-Digital Conversion,
- Restart the MCU if software is unable to periodically restart a WATCHDOG timer.

TIMERS

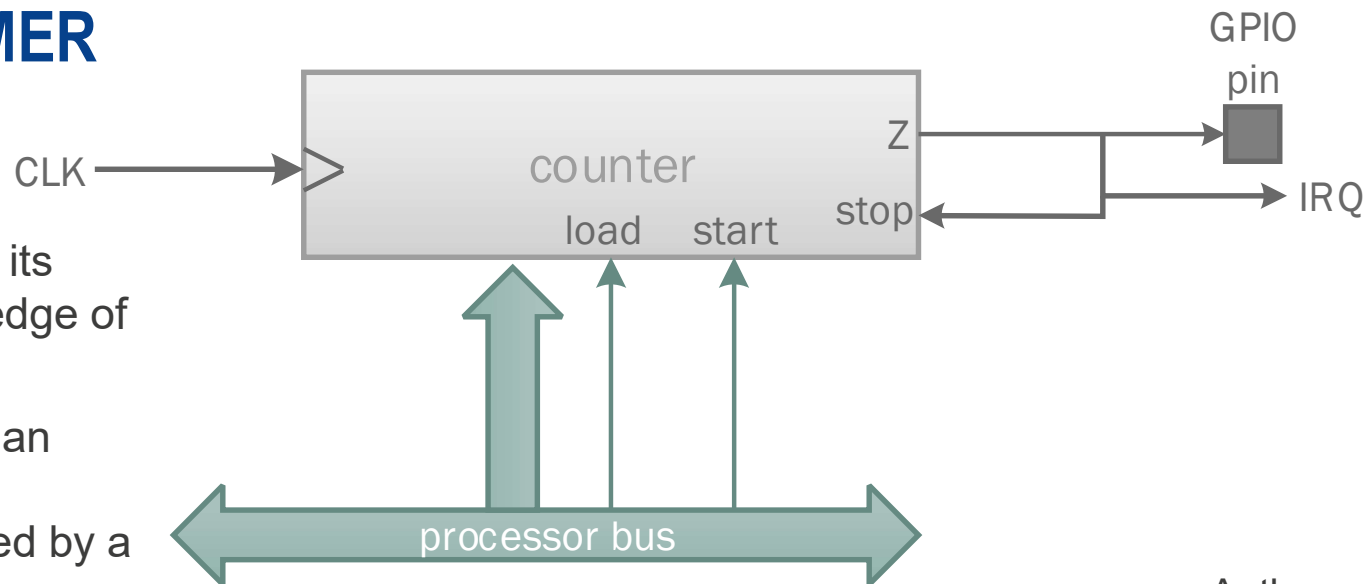
Timers/Counters vary in their characteristics, such as:

- edge of the input signal used for counting: positive, negative, or both edges;
- count direction: up, down, or both;
- what is being counted: clock pulses (timers) or other input signal (counters);
- periodic timers vs single-shot;
- a number of actions can be taken at the end of the timing period: change state of GPIO pin, generate IRQ, stop the timer, reload the timer;
- number of bits of the counter circuit: typically 32 bits for ARM MCUs;
- since the frequency of the input signal and number of bits of the counter determine the maximum timing period, a prescaler may be used to reduce the input frequency.

A VERY SIMPLE TIMER

Description:

- The counter decrements its value on every positive edge of the CLK.
- The processor can write an initial count value by performing a load followed by a start command.
- When the count reaches 0 the Z output is activated.
- The Z output may command a change to the GPIO pin, generate an Interrupt Request (IRQ), and stop (or reload) the counter.



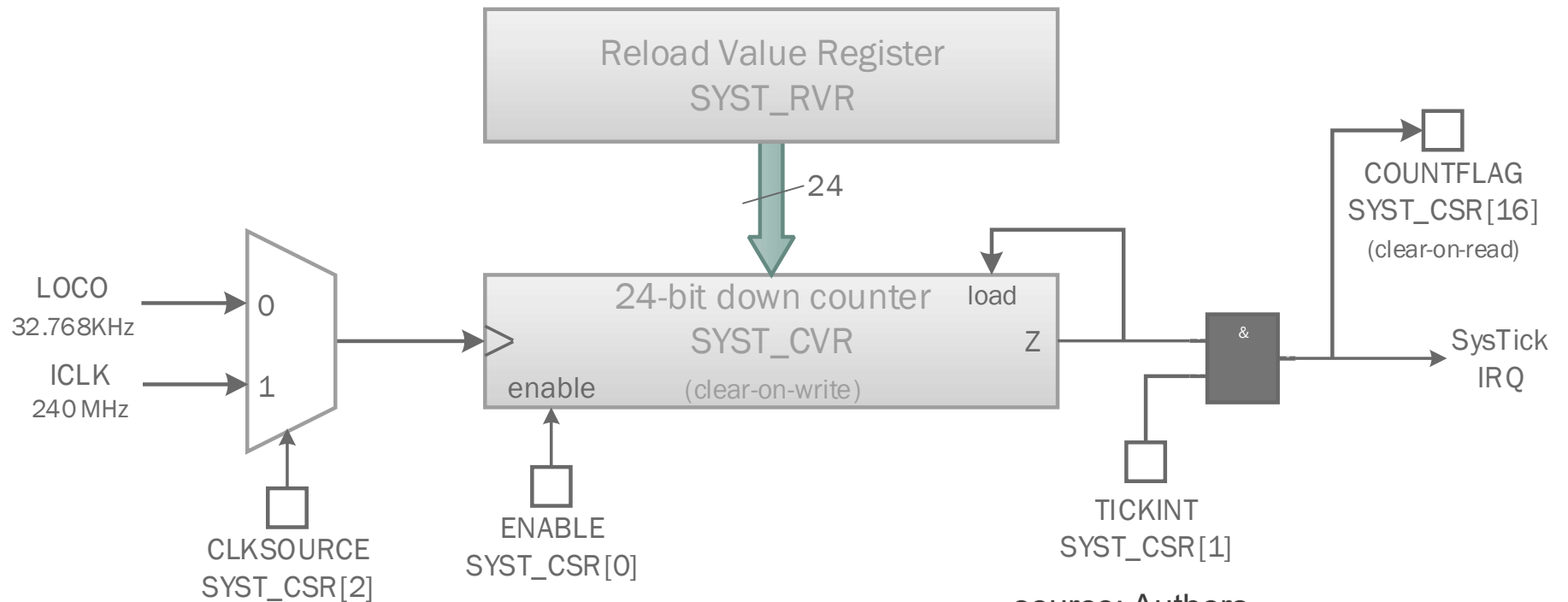
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TIMER CASE STUDY 1 - SYSTICK

- The SYSTICK is a timer available in all Cortex-M processors.
- It is a simple 24-bit counter with auto-reload.
- Its main use is to generate periodic interrupts required by most Embedded Operating Systems.
- Since its structure and operation is defined by ARM, its interface is standard, regardless of MCU supplier.

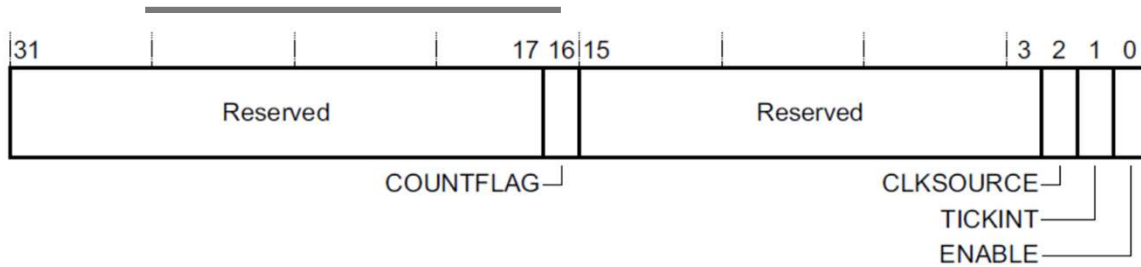
TIMER CASE STUDY 1 - SYSTICK

SysTick auto-reload has a period of $\text{SYST_RVR} + 1$ clock cycles.

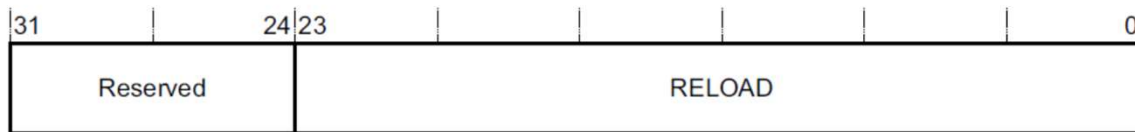


source: Authors
(based on ARM documentation:
DDI0403D ARMv7-M Architecture Reference Manual)

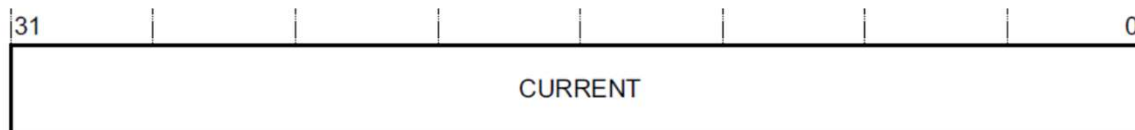
TIMER CASE STUDY 1 - SYSTICK



SYST_CSR controls the operation of SysTick. When this register is read, COUNTFLAG is cleared.



SYST_RVR holds the 24-bit reload value. Writing a 0 to this register disables SysTick.



SYST_CVR is the current count value. By writing any value to it the register is cleared.



SYST_CALIB values are factory defined. TENMS, if available, holds the reload value corresponding to 10 ms for the reference clock.

TIMER CASE STUDY 1 - SYSTICK

Timing diagram for the scenario where SYST_RVR holds the value 2.

As the reload occurs on the next positive edge of the clock, the SysTick period is of 3 clock cycles.

If enabled, every reload generates a SysTick interrupt request (IRQ).

